

Toward a Sustainable Future: Biofiber

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by

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The stock of fossil fuels is limited, but the risk is limitless. So, eliminating petroleum-based products from daily life is not only difficult but also not impossible. Keeping in mind the issue of environmental pollution, people are turning away from petroleum-based products towards bio-based products. They are searching for 'sustainable' solutions, i.e., renewable natural resources for regeneration and finding alternative pathways. It will be long-lasting, environmental friendly as well as providing sustainability to the environment and being beneficial to the humanity.

The demand for bio-materials made from renewable resources is now global. Scientists and technologists around the world are engaged creating highly useful products such as bio-based fuels, fibers, films, foams, and fabrics, whose main components are bio-molecules like cellulose, starch, or proteins. The goal is to create biodegradable and compostable products using adequate and sustainable use of natural resources, which will significantly reduce greenhouse gas emissions.

Bio-fibers obtained from renewable resources mainly are two types: plant and animal fibers. Plant fibers refer to natural fibers obtained from plants, such as cotton, jute, hemp, bamboo, flax, sisal, kenaf, pineapple leaves, etc. Plant biomass is first separated into monomers, and then these monomers are polymerized to create high molecular weight polymers through a polymerization process. Then, through appropriate spinning processes, these polymers are transformed into fibers. On the other hand, animal fibers include wool from sheep, silk from silkworms, etc., which have been used in the textile and fashion industry for centuries.

The demand for bio-fibers has increased due to awareness of circular economy practices and reducing carbon footprints. For a health conscious society, the significant demand for bio-fibers is evident. The popularity of bio-fibers in the textile, packaging, and fashion industries is rising. Many renowned brands are incorporating bio-fibers into their products. Bamboo fiber is used to create special types of socks, high-quality bathrobes, towels, and sanitary napkins with anti-bacterial properties. Hemp or cannabis plant fibers are used to create high-quality clothing fabrics. Jute fiber is used to make 'Sonali Bags,' which serve as alternatives to non-biodegradable plastic bags. Companies like Pact, No Nasties, and Grassroot are bringing fashionable dresses, sanitary napkins, and everyday essentials bags into the market, utilizing bio-fibers. Various startups are working on new ideas using plant-based fibers both nationally and internationally.

The European Union and other developed countries have long shifted their focus towards bio-based fibers, imposing restrictions on petroleum-based synthetic fibers. Despite of having various bans on the use of petroleum-based products worldwide, countries like India still witness extensive usage of plastic bags in the market. It is long back that the plastic bags below 50 microns were prohibited in this country. But the use of it in the markets has been so rampant that now we are at the point of backtracking. We keep mum, like silent spectators, let alone raise a protest against this malpractice. Ours are nuclear families in theory, but we do not at all live upto this ideal. For we use separate bags for separate vegetables. What an irony! The question also arises how the very coffee cups in which lies the plastic lining, can be biodegradable. Perhaps in the days ahead we'll cease to exist while in search of light in the womb of darkness or darkness in the womb of light, and then, to be true, though hard a reality, we'll have no scope at all to raise the slogan of not being "sustainable".